# A proof of the conjectured recurrence for OEIS A226302

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**Abstract**

OEIS A226302 carries a conjectured linear recurrence with polynomial coefficients, of order 5. The entry separately states a recurrence of order 4 that is not labelled a conjecture. We prove that the first is a consequence of the second, so the conjecture holds. The argument is a division in the Ore algebra $\mathbb{Q}(n)[N]$, where $N$ is the shift $Nf(n) = f(n+1)$: writing $C$ and $P$ for the two recurrences read as operators, we exhibit $Q$ with $C = QP$, whence $C(a) = Q(P(a)) = Q(0) = 0$. No generating function is used. The division is exact rational-function arithmetic, and the finitely many $n$ excluded by denominators of $Q$ are computed rather than ignored.

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## 1 The sequence and the two recurrences

OEIS A226302 is "a(n) = P_n(-1), where P_n(x) is a certain polynomial arising in the enumeration of tatami mat coverings". It has offset 2 and begins

$$a(2), \dots, a(9) \;=\; 1, -1, 2, -4, 6, -14, 20, -48, \;\dots$$

The entry carries the following comment, still recorded as a conjecture:

> **Conjecture:** (-n+2)*a(n) +(-n+2)*a(n-1) +2*(3*n-11)*a(n-2) +2*(3*n-14)*a(n-3) +4*(-2*n+9)*a(n-4) +8*(-n+6)*a(n-5)=0. - *R. J. Mathar*, Nov 06 2013

As of the "Last modified" line on the live entry (2025-11-05, revision 30) it is still a conjecture, and no proof appears among the entry's links, formulas or programs.

The same entry also states, *not* as a conjecture:

> Recurrence (for n>5): (n-5)*(n-2)*a(n) = -2*(n-4)*a(n-1) + 2*(n-5)*(3*n-10)*a(n-2) + 4*(n-4)*a(n-3) - 8*(n-5)*(n-4)*a(n-4). - *Vaclav Kotesovec*, Jun 14 2015

This second relation is what the argument below rests on; it is taken as given, exactly as a posted generating function would be. It was checked against the entry's own DATA before being used, and holds on every published term it reaches.

## 2 Recurrences as operators, and what division proves

Let $N$ be the shift operator, $(Nf)(n) = f(n+1)$, acting on sequences. Polynomials in $n$ and $N$ generate the Ore algebra $\mathcal{A} = \mathbb{Q}(n)[N]$, in which multiplication is associative and distributive but not commutative: moving $N$ past a coefficient shifts it,

$$N\,q(n) \;=\; q(n+1)\,N.$$

A recurrence $\sum_{i=0}^{r} p_i(n)\, a(n-i) = 0$ becomes an element of $\mathcal{A}$ after re-indexing $n \mapsto n + r$, namely

$$L = \sum_{j=0}^{r} q_j(n)\, N^j, \qquad q_j(n) = p_{r-j}(n + r),$$

and the recurrence says exactly that $L$ annihilates the sequence, $L(a) = 0$.

**Lemma 1.** *Let $P, C \in \mathcal{A}$ and suppose $C = QP$ for some $Q \in \mathcal{A}$. If $P(a) = 0$ then $C(a) = 0$, at every $n$ where the coefficients of $Q$ are defined.*

*Proof.* $C(a) = (QP)(a) = Q\big(P(a)\big) = Q(0) = 0$. The only caveat is the one stated: a coefficient of $Q$ is a rational function of $n$, so the step is valid at every $n$ that is not a pole of one of them. □

**Lemma 2.** *Right division works in $\mathcal{A}$: given $C$ and $P \neq 0$, there are $Q, R \in \mathcal{A}$ with $C = QP + R$ and $\deg_N R < \deg_N P$.*

*Proof.* Induct on $\deg_N C$. If $\deg_N C < \deg_N P$ take $Q = 0$, $R = C$. Otherwise let $d = \deg_N C - \deg_N P$ and let $c$, $p$ be the leading coefficients of $C$ and $P$. By the commutation rule, the leading coefficient of $N^d P$ is $p(n+d)$, which is a nonzero element of $\mathbb{Q}(n)$ and hence invertible there. Then $C - \dfrac{c(n)}{p(n+d)} N^d P$ has strictly smaller degree in $N$, and induction applies. □

So the question "does the conjecture follow from the stated recurrence?" is decided by running the division and looking at the remainder. Nothing is approximated: the coefficients live in $\mathbb{Q}(n)$ and the arithmetic is exact.

# 3 The computation for A226302

Reading both relations as operators and dividing the conjectured one by the stated one leaves remainder 0, with quotient

$$Q = \left(-\frac{1}{n}\right) + \left(-\frac{1}{n}\right) N.$$

Hence $C = QP$ in $\mathcal{A}$.

The coefficients of $Q$ are rational functions of $n$. Their only integer poles are $n \in \{0\}$, all of them below the range claimed here, so no $n$ in question is excluded.

**Theorem 3.** *The conjectured recurrence*

$$(2-n)\, a(n) + (2-n)\, a(n-1) + 2\,(3n-11)\; a(n-2) + 2\,(3n-14)\; a(n-3) - 4\,(2n-9)\; a(n-4) - 8\,(n-6)\; a(n-5) =$$

*holds for every $n \geq 7$.*

*Proof.* By Lemma 1 applied to the factorisation above, using that the stated recurrence annihilates the sequence. The integer poles of the coefficients of $Q$ all lie below $n = 7$, so the lemma applies throughout the stated range. □

# 4 Verification

Every number above is an output of a script written separately from this note, and three independent checks were run.

First, the factorisation was not assumed but computed, and then confirmed by multiplying back: $QP$ was expanded in $\mathcal{A}$ and its coefficients cancelled against those of $C$ to zero, as rational functions of $n$.

Second, the relation the argument rests on was checked against the entry rather than trusted: the stated recurrence was evaluated on the published terms in exact integer arithmetic and holds on every one it reaches.

Third, the conjecture itself was evaluated the same way, directly and without reference to either operator: for each $n$ in range the sum $\sum_i p_i(n)a(n-i)$ was formed from the entry's own values and found to vanish, for all 30 values of $n$ from $n = 7$ upward.

## References


[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A226302.

[2] O. Ore, *Theory of non-commutative polynomials*, Ann. of Math. 34 (1933), 480–508.

[3] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011, Chapter 7.

[4] M. Bronstein and M. Petkovšek, *An introduction to pseudo-linear algebra*, Theoret. Comput. Sci. 157 (1996), 3–33.